Game of Registrars: An Empirical Analysis of Post-Expiration Domain Name Takeovers

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Every day, hundreds of thousands of Internet domain names are abandoned by their owners and become available for re-registration. Yet, there appears to be enough residual value and demand from domain speculators to give rise to a highly competitive ecosystem of *drop-catch* services that race to be the first to re-register potentially desirable domain names in the very instant the old registration is deleted. To pre-empt the competitive (and uncertain) race to re-registration, some registrars sell their own customers' expired domains *pre-release*, that is, even before the names are returned to general availability.

These practices are not without controversy, and can have serious security consequences. In this paper, we present an empirical analysis of these two kinds of post-expiration domain ownership changes.We find that 10% of all `com` domains are re-registered on the same day as their old registration is deleted. In the case of org, over 50% of re-registrations on the deletion day occur during only 30 s. Furthermore, drop-catch services control over 75% of accredited domain registrars and cause more than 80% of domain creation attempts, but represent at most 9.5% of successful domain creations. These findings highlight a significant demand for expired domains, and hint at highly competitive re-registrations.

Our work sheds light on various questionable practices in an opaque ecosystem. The implications go beyond the annoyance of websites turned into “Internet graffiti”, as domain ownership changes have the potential to circumvent established security mechanisms.

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